



# RIVER VALLEY HIGH SCHOOL

## YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE  
NAME

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CLASS

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CENTRE  
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### H1 CHEMISTRY

**8873/02**

Paper 2 Structured Questions

**12 September 2018**

**2 hours**

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

#### READ THESE INSTRUCTIONS FIRST

Write your name, class, centre number and index number on all the work you hand in.

Write in dark blue or black pen on both sides of paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

#### Section A

Answer **all** questions.

#### Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided. Do NOT write anything on it.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |    |   |         |   |    |   |       |     |       |
|--------------------|----|---|---------|---|----|---|-------|-----|-------|
| Question No.       | 1  | 2 | 3       | 4 | 5  | 6 | 7 / 8 | s.f | units |
| Marks              | 9  | 8 | 14      | 8 | 12 | 9 | 20    |     |       |
| Paper 1            | 30 |   | Paper 2 |   | 80 |   | Total | 110 |       |

This document consists of **24** printed pages.

## Section A

Answer all the questions in this section, in the space provided.

- 1 Magnesium, aluminium, **W** and **X** are four different elements in Period 3. The following are some of their properties:

The chloride of **W** dissolves in water to give a strongly acidic solution and the oxide of **W**.

The two oxides of **X** are solids and reacts with water to form acidic solutions of pH 2.

- (a) Identify the elements, **W** and **X**, and write the balanced equations for the reactions described above.

**W**: .....

Chloride of **W** dissolved in water (1 equation):

.....

**X**: .....

Two oxides of **X** reacting with water (2 equations):

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.....

[3]

- (b) Aluminium oxide is a white solid that is soluble in NaOH(aq) to give a colourless solution.

- (i) Write a balanced equation, with state symbols, to describe the reaction of aluminium oxide with NaOH(aq).

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[1]

- (ii) Describe the observations when HCl(aq) is carefully added to this colourless solution, until HCl(aq) is in excess.

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[2]

- (c) State and explain in terms of bonding and structure, the trend in volatility of the Group 17 elements.

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[2]

- (d) Explain why the 1<sup>st</sup> ionisation energy of aluminium is of lower magnitude compared to that of magnesium.

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[1]

[Total: 9]

- 2 Geckoes have developed hierarchical foot nanostructures that allow them to climb walls.

(a) Define the term “nanostructure”.

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[1]

(b) Describe and explain the wall-climbing ability of geckos.

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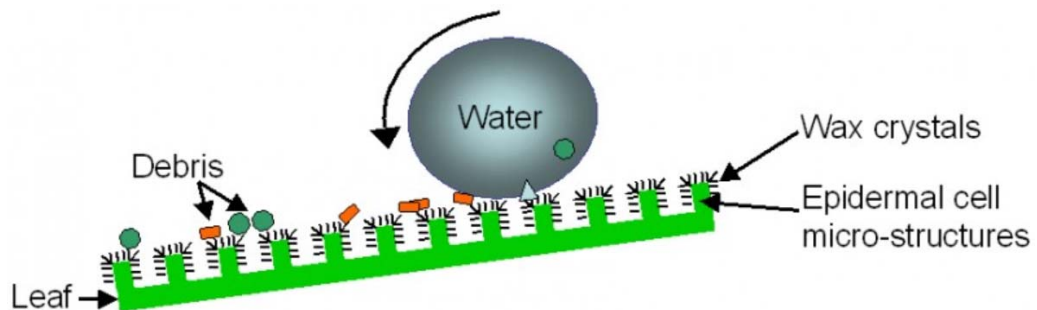
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[3]

- (c) Lotus effect describes the self cleaning properties of the lotus leaf due to its super hydrophobicity. There are wax crystals in nanoscale covering microstructures on the lotus leaf. The diagram below shows how water helps in the removal of debris.



Source: [http://urban-gallery.net/txl/?page\\_id=3336](http://urban-gallery.net/txl/?page_id=3336)

- (i) Given that debris on lotus leaf are mostly hydrophobic, and by considering the contact area between debris and the lotus leaf, suggest a reason for the ease of removal of debris on lotus leaf.

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[2]

- (ii) By considering the surface area to volume ratio, suggest and explain if the lotus leaf will still exhibit self-cleaning properties if a hydrophobic organic solvent is used in place of water.

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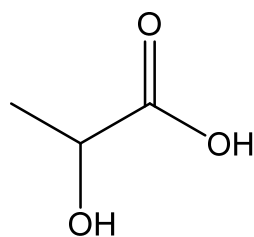
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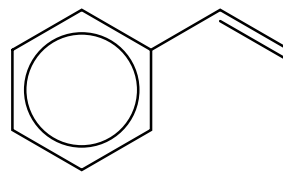
[2]

[Total: 8]

- 3 Lactic acid and styrene are very useful monomers in the synthesis of various polymers.



lactic acid



styrene

- (a) Lactic acid undergoes oxidation to form compound **A**.

State the reagents and conditions for the oxidation and draw the structure of compound **A**.

Reagents and conditions: .....

[2]

- (b) Lactic acid undergoes reduction with  $\text{LiAlH}_4$  to form compound **B**.

Draw the structure of compound **B**.

[1]

- (c) Lactic acid is the only monomer in the formation of polylactic acid (PLA).

Identify the type of polymerisation, state any reagents and conditions needed for the polymerisation.

Draw the structure of PLA, showing 2 repeating units.

Reagents and conditions: .....

Type of polymerisation: .....

[3]

- (d) Polystyrene is a thermoplastic.

- (i) Using a simple diagram, explain the structural difference between thermoplastic and thermosetting plastic.

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[3]

- (ii) State a property of polystyrene.

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[1]

- (iii) Identify the type of polymerisation and draw the structure of polystyrene, showing 1 repeating unit.

Type of polymerisation: .....

[2]

- (d) How does the recyclability of thermoplastic compare to that of thermosetting plastic?

Explain your answer.

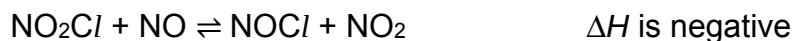
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[2]

[Total: 14]



- 4 Nitrosyl chloride,  $\text{NOCl}$ , can be formed between nitryl chloride and nitric oxide in the following reversible reaction:



- (a) A 3:1 molar ratio mixture of  $\text{NO}_2\text{Cl}$  and  $\text{NO}$ , at a total initial concentration of  $5.00 \text{ mol dm}^{-3}$ . When equilibrium was reached at time  $t_1$  at  $800 \text{ K}$ , the concentration of  $\text{NO}_2\text{Cl}$  was found to be  $2.90 \text{ mol dm}^{-3}$ .

Complete the table below.

Calculate the value of the equilibrium constant,  $K_c$ , of the reaction and indicate the unit of  $K_c$ .

|                                                    | $\text{NO}_2\text{Cl}$ | + | $\text{NO}$ | $\rightleftharpoons$ | $\text{NOCl}$ | + | $\text{NO}_2$ |
|----------------------------------------------------|------------------------|---|-------------|----------------------|---------------|---|---------------|
| Initial concentration/<br>$\text{mol dm}^{-3}$     |                        |   |             |                      | 0             |   | 0             |
| Change in concentration/<br>$\text{mol dm}^{-3}$   |                        |   |             |                      |               |   |               |
| Equilibrium concentration/<br>$\text{mol dm}^{-3}$ | 2.90                   |   |             |                      |               |   |               |

[3]

- (b) At time  $t_2$ , more  $\text{NO}_2\text{Cl}$  was introduced into the vessel at 800 K, causing the concentration of  $\text{NO}_2\text{Cl}$  to increase to  $3.25 \text{ mol dm}^{-3}$ . The system was allowed to reach equilibrium before the temperature was increased to 1000 K at  $t_3$ . A new equilibrium was established at  $t_4$ .

In the concentration-time axes below, sketch the graph that would be observed from time = 0 to time =  $t_4$ , clearly indicating the values of the concentration of  $\text{NO}_2\text{Cl}$  at time = 0,  $t_1$  and  $t_2$ .



[2]

- (c) Given that nitrogen is the central atom, draw the dot-and-cross diagram for  $\text{NOCl}$ .

[1]

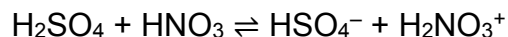
- (d) State the shape and bond angle of the  $\text{NOCl}$  molecule.

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[2]

[Total: 8]

- 5 (a) Sulfuric acid and nitric acid react according to the following equation:



With reference to the Brønsted-Lowry theory and the chemical species in the equation above, explain the terms **acid**, **base**, **conjugate acid** and **conjugate base**.

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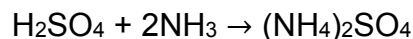
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[2]

- (b) In an acid-base titration, 25.0 cm<sup>3</sup> of 0.0210 mol dm<sup>-3</sup> H<sub>2</sub>SO<sub>4</sub>(aq) was titrated with 0.0500 mol dm<sup>-3</sup> NH<sub>3</sub>(aq) from a burette. The equation for the reaction is shown below:



- (i) Determine the volume of NH<sub>3</sub>(aq) required to reach equivalence point.

[1]

- (ii) Calculate the number of moles of H<sub>2</sub>SO<sub>4</sub> remaining after 10.00 cm<sup>3</sup> of NH<sub>3</sub> has been added. You may assume that H<sub>2</sub>SO<sub>4</sub> behaves as a strong diprotic acid.

[2]

- (iii) Hence, calculate the pH of the solution obtained after 10.00 cm<sup>3</sup> of NH<sub>3</sub> has been added.

[2]

- (c) An alkaline buffer can be obtained by preparing a solution containing both a weak base and its salt. The pH of such solution can be found by determining its pOH value, which is given by the Henderson-Hasselbalch equation below:

$$\text{pOH} = -\lg(K_b) + \lg\left(\frac{[\text{salt}]}{[\text{weak base}]}\right)$$

- (i) Write an expression for the  $K_b$  of NH<sub>3</sub>.

[1]

- (ii) Given that the  $K_b$  value of NH<sub>3</sub> is  $1.8 \times 10^{-5}$ , calculate the mass of NH<sub>4</sub>Cl(s) that needs to be added to 1.00 dm<sup>3</sup> of 0.0500 mol dm<sup>-3</sup> NH<sub>3</sub>(aq) to obtain an alkaline buffer with a pH of 9.5.

You may assume that the volume of the solution remains constant.

[2]

(d) One important application of buffers is the biological buffer in blood which keeps the pH of blood relatively constant.

(i) State the two chemical species that constitute blood buffer.

..... [1]

(ii) Explain, with the aid of an equation, how the blood buffer keeps the pH of blood relatively constant when cellular respiration produces lactic acid.

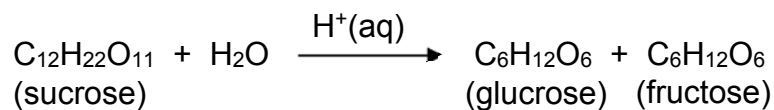
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..... [1]

[Total: 12]

- 6 (a)** Sucrose, or table sugar, is the most common natural food sweetener. In acidic medium, sucrose is readily hydrolysed to a mixture of glucose and fructose. The reaction is also acid-catalysed.



A series of experiments was carried out at room temperature to investigate the kinetics of this reaction, using  $0.850 \text{ mol dm}^{-3}$  sucrose solution and  $1.23 \text{ mol dm}^{-3}$  hydrochloric acid.

| Expt | Volume of sucrose / $\text{cm}^3$ | Volume of $\text{HCl}$ / $\text{cm}^3$ | Volume of water / $\text{cm}^3$ | Initial rate of reaction / $\text{mol dm}^{-3} \text{min}^{-1}$ |
|------|-----------------------------------|----------------------------------------|---------------------------------|-----------------------------------------------------------------|
| 1    | 20                                | 30                                     | 0                               | $1.77 \times 10^{-3}$                                           |
| 2    | 20                                | 20                                     | 10                              | $1.18 \times 10^{-3}$                                           |
| 3    | 10                                | 30                                     | 10                              | $8.85 \times 10^{-4}$                                           |
| 4    | 40                                | 20                                     | 10                              | ?                                                               |

- (i) Using the data given above, determine the order of reaction with respect to sucrose and HCl.

[illegible]

[2]

- (ii) Hence, write the rate equation and deduce the rate constant of the reaction, stating its units.

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[2]

- (iii) Using your answer in (a)(ii), calculate the initial rate of reaction for Experiment 4.

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[2]

- (b) In bacteria, sucrose is broken down into glucose and fructose by the enzyme, invertase. Experiments were done to measure the rate of the enzyme-catalysed hydrolysis reaction for different concentrations of sucrose.

- (i) Sketch a graph to show how the rate of this enzyme-catalysed hydrolysis reaction varies with the concentration of sucrose until sucrose is in large excess.

[1]

- (ii) Explain the shape of the graph in (b)(i). In your answer, make reference to the order of reaction with respect to sucrose.

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[2]

[Total: 9]



**Section B**

Answer **one** question from this section, in the spaces provided.

- 7 (a) Using s,p,d,f notations, write the ground state electronic configuration for aluminium.

Account for the significant increase in ionisation energy when the fourth electron is removed.

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[3]

- (b) Using s,p,d,f notations, write the ground state electronic configurations for silicon.

The first ionisation energy of aluminium is smaller compared to that of silicon.  
With reference to the electronic configurations, account for these observations.

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[4]

- (c) The first ionisation energies of boron, aluminium and gallium decreases down the Group.

Account for these observations.

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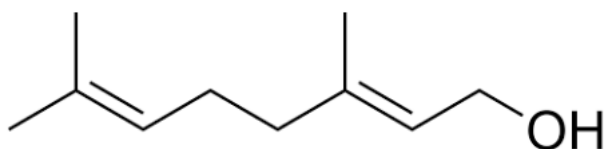
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[2]

- (d) Organic chemicals used in fragrance and flavouring are carefully examined to ensure safety of the public. Many of such chemicals are synthesised based on natural compounds isolated from various sources. Geraniol is one of the chemicals that gives the fruity fragrance in basil and lemongrass.



geraniol

Other than oxidation, suggest a chemical test to confirm the identity of geraniol.

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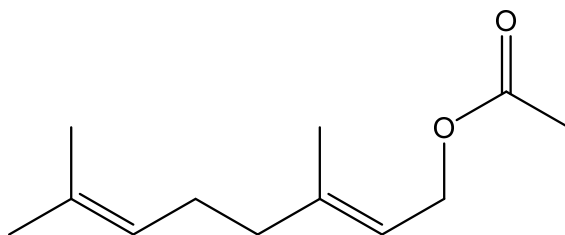
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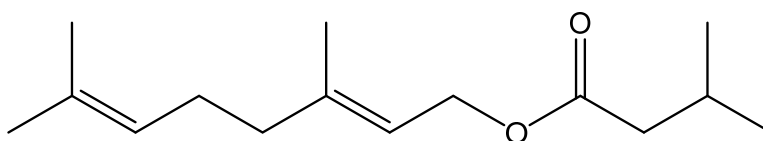
- (e) Calculate the percentage by mass of carbon in the geraniol molecule.

[2]

- (f) A mixture of geranyl acetate and geranyl isovalerate undergoes reaction 1 to give geraniol and 2 other products.



geranyl acetate



geranyl isovalerate

- (i) State the reagents and conditions for reaction 1.

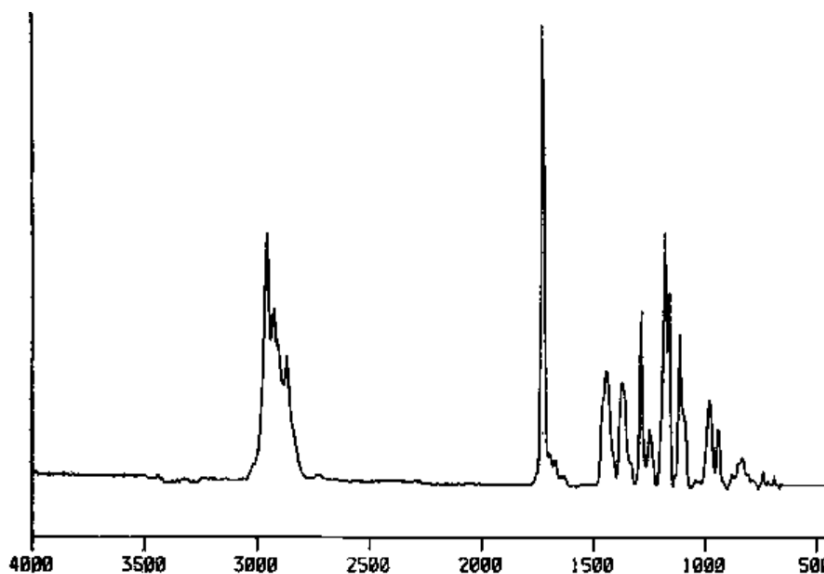
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[1]

- (ii) Draw the full structural formulae of the 2 other products of reaction 1.

[2]

- (iii) The infra-red spectrum of geranyl isovalerate is shown below.



Source: <http://www.fao.org/docrep/w6355e/w6355e16.htm>

Using the *Data Booklet*, identify an infra-red absorption peak that is shown in geranyl isovalerate but not geraniol.

..... [1]

- (iv) Using the *Data Booklet*, identify one of the infra-red absorption ranges that will be shown by geraniol but not geranyl isovalerate and geranyl acetate.

..... [1]

- (v) In an experiment, a sample of geranyl acetate was vaporised, ionised and passed through an electric field. Analysis revealed that a beam of  $(\text{CH}_3)_2\text{CCHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCH}_2\text{OCOCH}_3^+$  ions gives an angle of deflection of  $20^\circ$ .

Calculate the angle of deflection for a sample of singly charged geranyl isovalerate ions,  $(\text{CH}_3)_2\text{CCHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCH}_2\text{OCOCH}_2\text{CH}(\text{CH}_3)_2^+$

[2]

[Total: 20]

- 8 The bromate(V) ion,  $\text{BrO}_3^-$ , is a type of bromine oxyanion. It has been used in hair straightening products due to its strong oxidising power. However, accidental ingestion of these hair products can result in oxidative damage to nerve cells.

(a) In an experiment, a student carried out the following procedure to investigate the oxidation of  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$  by  $\text{BrO}_3^-$ .

1. Weigh accurately 3.420 g of solid  $\text{FeSO}_4$  in a clean and dry 100  $\text{cm}^3$  beaker.
2. Using a burette, add 25.00  $\text{cm}^3$  of dilute  $\text{H}_2\text{SO}_4$  into the beaker to dissolve the solid.
3. Transfer the resultant solution to a volumetric flask and top up to the 250  $\text{cm}^3$  mark using deionised water.
4. Shake the flask well and label the resultant solution **FA 1**.
5. Pipette 25.0  $\text{cm}^3$  of **FA 1** into a conical flask and titrate it with 0.044  $\text{mol dm}^{-3}$   $\text{NaBrO}_3$  solution from a burette.
6. Repeat the titration as many times as required to obtain reliable results.

(i) Given that  $\text{NaBrO}_3$  is colourless, suggest whether an indicator is required for the titration.

Explain your answer.

[1]

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 .....

(ii) Calculate the number of moles of  $\text{Fe}^{2+}$  in 3.420 g of  $\text{FeSO}_4$  and hence, determine the concentration of  $\text{Fe}^{2+}$  in **FA 1**.

[2]

(iii) Calculate the number of moles of  $\text{Fe}^{2+}$  in the 25.0  $\text{cm}^3$  of **FA 1** pipetted into the conical flask for each titration.

[1]

- (iv) The following table shows the titration results of the student.

| titration number                          | 1     | 2     |
|-------------------------------------------|-------|-------|
| initial burette reading / cm <sup>3</sup> | 0.00  | 12.50 |
| final burette reading / cm <sup>3</sup>   | 25.60 | 38.20 |

Determine the average volume of NaBrO<sub>3</sub> required to react with 25.0 cm<sup>3</sup> of **FA 1**.

Hence, calculate the number of moles of BrO<sub>3</sub><sup>-</sup> required to react with the number of moles of Fe<sup>2+</sup> obtained in **(a)(iii)**.

[2]

- (v) Calculate the number of moles of electrons gained by one mole of BrO<sub>3</sub><sup>-</sup>, leaving your answer to the nearest integer.

Determine the final oxidation state of bromine in the product.

[2]

- (vi) Given that the product is a singly-charged bromine oxyanion, suggest its chemical formula.

Hence, write a balanced equation for the reaction between BrO<sub>3</sub><sup>-</sup> and Fe<sup>2+</sup>.

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[2]

- (b) (i) Draw the dot-and-cross diagram for  $\text{BrO}_3^-$ .

[1]

- (ii) State the shape of  $\text{BrO}_3^-$  and the O–Br–O bond angle.

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[2]

- (iii) Using concepts from the Valence Shell Electron Repulsion theory, explain why  $\text{BrO}_3^-$  has a larger bond angle than water?

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[1]

- (iv) The table below shows the melting points of bromine-containing substances.

| substance          | melting point / °C |
|--------------------|--------------------|
| Br <sub>2</sub>    | -7.2               |
| BrO <sub>2</sub>   | 0                  |
| NaBr               | 747                |
| NaBrO <sub>3</sub> | 381                |

In terms of structure and bonding, explain why

- Br<sub>2</sub> has a lower melting point than NaBr;
- Br<sub>2</sub> has a lower melting point than BrO<sub>2</sub>;
- NaBrO<sub>3</sub> has a lower melting point than NaBr.

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[4]

- (c) Explain the conditions of temperature and pressure required for gases to be liquefied.

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[2]

[Total: 20]

- End of Paper -